PHY 210: Mathematical Methods of Physical Sciences and Engineering

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One of the great strengths of a modern physics background is that it allows you to address problems in an enormous range of fields, from physics itself to economics, biology, ecology, computer science, complex systems, physical chemistry, geology, and engineering. The primary objective of this course is to provide a systematic introduction to mathematical techniques that will serve you both in future physics courses and in modeling interesting problems in your domain of choice. A significant secondary objective is to develop a level of mathematical sophistication that will allow you to confidently and competently explore further material on your own.

Concretely, by the end of this course you will be able to:

1. Describe a physical scenario mathematically (most often as a differential equation), solve the resulting model, verify your solution, and interpret it back in the real world (including assessing where the model breaks down).
2. Apply core tools of mathematical physics: ordinary differential equations, Laplace transforms, complex numbers, series approximations, linear algebra (including matrix manipulation, eigenvalues and eigenvectors) and normal modes, multidimensional integrals, vector calculus, Fourier series, and introductory partial differential equations.
3. Check your work the way scientists and engineers do: verify units, plug solutions back into the original equations, test limiting cases.
4. Use Python to visualize and explore mathematical models.

Instructor: Michael Lerner
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Class: MWF 9:25–10:40, McConnell 404
First meeting Wed Sep 9
Last meeting Mon Dec 14

Office hours: **TO BE SET** after the week-1 scheduling poll. I also have an open-door policy: stop in whenever my door is open. That's most of the time.

Text: Felder & Felder, *Mathematical Methods in Engineering and Physics* (some sections and all answers to odd problems are free at `https://www.felderbooks.com/mathmethods/`)

Website: Moodle
Python: `https://posit.smith.edu` – nothing to install or buy

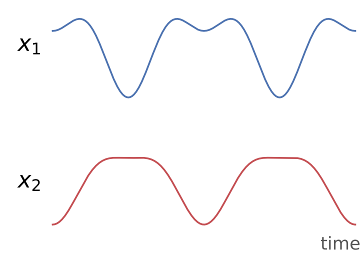

Figure 1: Two masses connected by springs. Each one's motion looks complicated, but it is secretly just two cosines added together. By week 9 you will be able to predict this motion exactly (it is a worked example in Chapter 6), and in week 15 it becomes the wave equation.

Prerequisites

MTH 212 (multivariable calculus) and PHY 111, 117, or 119, or permission of the instructor.

Course calendar

The complete day-by-day schedule (topics, reading due, homework due) is posted on Moodle and kept up to date there.

Weeks	Topics	Reading
1–3	Ordinary differential equations: setting them up from physical situations, arbitrary constants, separation of variables, guess and check, superposition	Ch. 1
3–4	Heaviside, Dirac delta, and the Laplace transform (with Python)	10.2, 10.10–11
5	Complex numbers and Euler's formula	Ch. 3
6–7	Linear approximations, Maclaurin and Taylor series	Ch. 2
7–9	Linear algebra: matrices as transformations, eigenvalues and eigenvectors, coupled oscillators and normal modes	Ch. 6
10–11	Integrals in two and more dimensions; cylindrical and spherical coordinates; line and surface integrals	Ch. 5
11–13	Vector calculus: gradient, divergence, curl – **defined geometrically, following Feynman** – then the divergence theorem, Stokes' theorem, and conservative fields	Feynman II 2–3; Ch. 8
14	Fourier series	Ch. 9
15	Partial differential equations: the heat equation, separation of variables, normal modes of the wave equation; Fourier transforms to close us out	Ch. 11; 9.6

A WORD ABOUT FEYNMAN. For the vector-calculus unit, Chapters 2 and 3 of *The Feynman Lectures on Physics, Volume II* are *assigned reading*: we will meet divergence and curl through Feynman's physical, geometric definitions before we see the formulas in Felder.[1]

How the course works

Everything in this course is worth points, and the course total is exactly 1000 points. Within each category, each assignment is weighted equally.

Key dates

Fri Sep 18	Math pretest due
Fri Sep 25	Quiz 1 (Ch. 1)
Fri Oct 9	Flex day
Mon Oct 12	No class (recess)
Fri Oct 23	Quiz 2 (Ch. 10, 3, 2)
Mon Oct 26	Non-Newtonian due
Fri Nov 20	Quiz 3 (Ch. 6, 5)
Wed Nov 25	No class
Fri Nov 27	No class
Dec 19–22	Final exam period

The flex day absorbs Mountain Day, snow, or wherever we're behind.

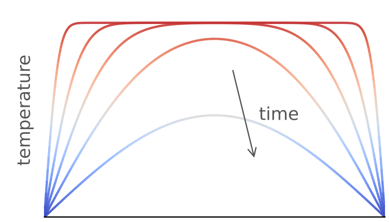

Figure 2: A uniformly hot bar with its ends held cold, relaxing toward equilibrium; color shows temperature. The sharp corners disappear almost instantly — by week 15 you'll know why.

[1] This part of Feynman is extremely conversational and readable, and you can read the whole thing free, with beautiful typesetting, at `https://www.feynmanlectures.caltech.edu/`.

Category	Number	Drop	Each	Total
Participation & pre-class check-ins (PCCIs)	39	4	1	35
Weekly homework (WHW)	13	1	25	300
Non-Newtonian Scientist	1	0	25	25
Quizzes (in class)	3	0	150	450
Final exam	1	0	190	190
Course total				**1000**

Grade scale (not curved)

A	93+	C+	77–79.9
A-	90–92.9	C	73–76.9
B+	87–89.9	C-	70–72.9
B	83–86.9	D+	67–69.9
B-	80–82.9	D	63–66.9
		D-	60–62.9
		E	below 60

Pre-class check-ins (PCCIs)

Most class days there will be a short PCCI due at the start of class, on paper. Many of these are the textbook's "Discovery Exercises," designed to introduce you to a topic *before* we cover it; others are short problems on material we've just covered. **You will receive full credit for a good-faith effort that includes a clear description of anything you are stuck on.** An incorrect or incomplete answer *without* such an explanation won't receive full credit.[2] A PCCI should take you less than 15 minutes; if one doesn't, please tell me. Turning in the day's PCCI is also how attendance is recorded, and your four lowest participation days are dropped, so occasional absences and off days cost you nothing.

[2] For example, if the question were to differentiate $f(g(x))$: "I don't know" earns nothing; the correct answer earns full credit; and so does "I think it's $f'(g(x))$, except I know the chain rule says to do something different because it's $f(g(x))$ and not just x – I'm not clear on what, though."

Weekly homework (WHW)

This is basically a course in problem-solving techniques. The most common and the most damaging mistake a student can make in such a course is to yield to the temptation to try and master the material by reading the text and listening to the lectures while working a minimum of problems. While reading and listening can give you valuable new ideas, problem-solving skills are only efficiently developed by solving as many problems as possible. For most students the understanding and knowledge gained from this course will be in direct proportion to the number of problems which they successfully complete.

Each week I will post a WHW problem list, organized in three tiers: **Warm-up**, **Essentials**, and **Depth**. I recommend starting with the warm-ups. If they're too easy, move on to Essentials. If you're interested in the Depth content, you can add those problems as well. *You are not expected to do all the problems.* I have included many so that if you feel you need more practice with a topic, you have lots of options. Do whichever problems you find you need practice on. If it starts to be too easy, that's a good sign that you're comfortable with it! If it remains challenging, reach out to me or go to tutoring hours for some help.[3]

[3] Answers to odd problems are in the book's Appendix M, free at `https://www.felderbooks.com/mathmethods/`.

Most weeks, one problem on the list is **required and computational** (Python, on `https://posit.smith.edu`); everything else is yours to choose from.

Each Friday by 10:00 PM, you will turn in on Moodle: (1) a photograph of your written work on **at least one problem** from that week's list, with a sentence or two about where you got stuck or what you'd like feedback on, and (2) your answers to a few short reflection questions, typed directly into the submission box. WHWs are graded on the same good-faith standard as PCCIs: thoughtful engagement earns full credit. Every deadline in this course lives on the Moodle calendar.

Quizzes

Three quizzes, in class, taking the full period, on the Fridays listed in the margin above. Sample quizzes with solutions will be posted beforehand, and quiz questions will be closely aligned with the in-class problems and the WHW lists. Quizzes are closed-book; you may bring **one 8.5×11 sheet of handwritten notes, both sides**.[4] Each quiz covers material from earlier weeks, never from the week the quiz is given, and I will announce the exact section coverage in class about a week ahead. If you need accommodations, contact me well before the quiz.

[4] You may use a basic calculator; no phones, laptops, or other internet-connected devices. Students with documented accommodations will, of course, have those accommodations met.

Final exam, with redemption problems

The final has two parts. The *required* part covers the last third of the course (vector calculus, Fourier series, and PDEs). The rest consists of *optional redemption problems* drawn from the three quizzes: if you do better on a redemption problem than you did on the corresponding quiz problem, the new score replaces the old one. There is no penalty for attempting a redemption problem – you cannot lose points. Do the required problems first; then do as many redemption problems as you want and have time for.

Partial credit, and how to get it

Several of the problems in this class are quite challenging. For the harder problems, my goal is to have you make the strongest possible effort toward **understanding** the solution. Thus, if you cannot fully solve a problem (on a quiz or the final), say whatever you can about the way in which a solution would proceed from where you stop; say whatever you can about the qualitative behavior of a solution; say whatever you can about the physical meaning of the solution. And always, always: **check your units**, and **plug your solution back into the original equation**.[5]

[5] You are never done solving a differential equation until you have verified that your solution works.

Late policy

It is extremely hard to catch up on late work (usually things are late because you are busy, and people don't tend to get less busy!). That is why every category has built-in drops: four participation days, one full WHW. My strong suggestion, when life happens, is to use the drops and prioritize getting your schedule back under control.

Beyond the drops: all students start the term with **3 free late passes**. You can spend a late pass to extend a WHW (or the Non-Newtonian assignment) to the following class period with no penalty. Late passes are not automatic – email me before the deadline to say you're using one, and once declared it is spent whether or not you turn the work in by the extended date.[6]

[6] Late passes cannot be applied to PCCIs (their whole value is priming you for that day's class – the drops cover them), to quizzes, or to the final. After your free passes are spent, further extensions cost 10%, then 20%, and so on, of that assignment's points.

The Non-Newtonian Scientist

[TODO: Rework this whole section. The two paragraphs below are repurposed diversity-statement prose from the Earlham syllabus, not Non-Newtonian assignment prose, and the one-sentence description of the assignment is a placeholder written before the actual prompt was located.]

I can hardly overstate how much inclusion, equity and diversity considerations have shaped my entire thinking around teaching and practicing physics. It affects everything from the way that physics is framed historically to the way that it is typically taught in your classrooms. My main goal is to create an environment that tells all of us that we can do physics; that invites people into physics who have never been invited, or who have been explicitly excluded. In this class, the "Non-Newtonian Scientist" assignment (due Monday, October 26) engages directly with who does physics and how the standard physics canon got assembled; the full prompt will be posted on Moodle.

Physics cannot be divorced from the world in which we study it. I expect that we will recognize that in this class, and I expect that we will create an inclusive and welcoming environment. More specifically, I expect that we will strive *not* to create an environment that excludes people. This is hard work. We will get things wrong. You will get things wrong. *I* will get things wrong. Especially when I get things wrong, I encourage you to tell me – directly, or through any of the anonymous channels below.

Use of artificial intelligence

AI is a powerful tool, and it offers real potential to enhance your learning in this course. Used improperly, it can genuinely hinder your development as a physicist. You are welcome to use AI in this class, provided you always adhere to the Honor Code: **using AI to solve homework, PCCI, quiz, or exam problems is a violation of the Honor Code and will be treated as such**, and any AI use in work you submit must be cited, like any other source.[7]

[7] Adapted, with thanks, from Will Raven.

Two things worth noting clearly: First, AI models make subtle errors, hallucinate, and hand you technically-correct solutions that teach you nothing; your job is to verify, question, and ultimately understand the physics. Second your PCCIs and WHWs are graded on good-faith effort, not correctness. There are no points available for a polished answer. The only thing being graded is the record of *your* thinking, including where you got stuck; having an AI polish that doesn't help your grade at all, and cheats you of learning.

We're happy to use AI where it actually helps in this class, but we have to be careful not to use it before then. So, feel free to use AI to dig further in, to get alternate explanations, to check rote algebra after you've done it, to explore new topics, etc. But keep it in check to make sure you're in charge of your own learning. Early in the semester, we will spend part of a class deciding *together* how AI should and shouldn't fit into homework in this course — computational work especially — and we'll post what we decide on Moodle next to this policy.

Policies and logistics

COMMUNICATION. My preferred mode of communication is email. I will check my email multiple times per day during the work week. If you have sent me an email and have not heard back from me within 24 hours (excluding weekends), you should email me again or drop by my office to ask your question in person. I cannot guarantee that I will regularly check my email on evenings or on weekends.[8]

[8] I expect that you will check your Smith email at least once per day: 24 hours after I email the class, I expect everyone has read it. Emailed assignments, due dates, and policy changes carry the same weight as this syllabus, and the same goes for Moodle.

WORKLOAD. This course should take about 12 hours of your time per week: about 4 in class, and about 8 outside it on reading, PCCIs, homework problems, and review. **If you find yourself spending significantly more (or less) than this, please let me know ASAP! That's an instructor-error that I can easily fix.**

SENSITIVE ISSUES. What if you have something you want to talk about, but don't feel comfortable doing so in a public (or non-anonymous) space? This often comes up with issues of racism, sexism, and other kinds of discrimination. It can come up with issues that relate to my behavior, the behavior of your classmates, or the world in general. Please feel *encouraged* to use either the anonymous feedback form linked on Moodle, or to leave a written note under my door.[9] Of course, if you feel comfortable and safe doing so, you are also encouraged to talk publicly or non-anonymously about these things! If you are not comfortable with any of these options, the Resources section below lists offices that can help, including confidential ones.

[9] If you want to write a note, you may want to type it up and print it out so that your handwriting is further anonymized.

RECORDING. My plan is to record from my iPad to both PDF and videos, posted to Moodle. We'll see how technology works.

HONOR CODE. **[TODO: check this text against the current Smith Honor Code language before distribution]** Students and faculty at Smith are part of an academic community defined by its commitment to scholarship, which depends on scrupulous and attentive acknowledgement of all sources of information and honest and respectful use of college resources. Smith College expects all students to be honest and committed to the principles of academic and intellectual integrity in their preparation and submission of course work and examinations. All submitted work of any kind must be the original work of the student, who must cite all the sources used in its preparation (including, for example, any AI tools). Students voted to establish the academic honor system in 1944. The basis of the Academic Honor Code is articulated in Article X of the SGA Constitution and Article VII of the SGA Bylaws.[10]

[10] In this class specifically: you are encouraged to work together on the WHW problems and to study together for quizzes, but PCCIs should be attempted individually first, everything you turn in must be your own work, and quizzes and the final must be done individually.

ACADEMIC ACCOMMODATIONS. If you need classroom accommodations, please contact me and include your letter from the Accessibility Resource Center (ARC, arc@smith.edu). I promise to be flexible in meeting any documented need for accommodations in a way that works well for you. If you would like some accommodation but do not have an ARC letter, let me know and we can discuss your concerns: with or without a letter, I want to provide you the support you need to succeed in this course.

LAND ACKNOWLEDGMENT. We acknowledge that we are on Indigenous land: the territory of the Nipmuc people. We are grateful for the opportunity to live, learn, and grow on this sacred land, and extend our respect to citizens of this nation who live here today, and to their ancestors who have lived here for hundreds of generations. We recognize the repeated violations of sovereignty, territory, and water perpetrated by invaders who have impacted the original inhabitants of this land for 400 years. We know this acknowledgement is insufficient, and does not undo the harm that has been done and continues to be perpetrated now against Indigenous people and their land and water.

COURSE FEEDBACK. During the last week of classes, approximately 15 minutes at the beginning of one class will be set aside for you to complete the course feedback questionnaire.

Advice from former students

Gary Felder asked his Spring 2021 math-methods students what advice they would give to future students. Their answers are reproduced verbatim below (their course used Mathematica where we use Python).

- Go to tutoring and office hours and ask questions whenever you are confused. Most importantly do the readings.
- It's a good course, but it is challenging. Make sure you read the textbook, and go to tutoring/office hours if you're unsure about anything.
- Definitely go to drop–in hours if you can, and studying with other people can be very helpful.
- This course will show you how many of the seemingly tangential bits of math you have learned in earlier courses are very useful for solving problems and how they are often linked to a far richer set of mathematical skills/concepts. Due to this, it is really important to be comfortable with the concepts from earlier courses that are embellished in this course(luckily Gary provides an intro test that allows you to make sure you have the required concepts down). By the same logic, it is really important to make sure you understand each chapter of the course, because most subsequent sections build on the tools you acquire in each unit.
- Start your weekly assignment early so that you have time to go to Office Hours for help! Make sure you understand every homework problem and are able to apply the same concepts to similar problems.
- Go to office hours. Don't spend hours on the homework because you can always go to office hours and that will be a more effective use of your time.
- I would tell the student that even though "math methods" sounds scary, it's not that bad. The professor and the weekly homework prepare you for the exams. Always make sure to check your work and units.
- I would tell students to really take advantage of office hours and tutor hours especially when it comes to understanding and using Mathamatica because it will save you so much time and make your understanding of the course so much better.

- Don't take this class unless you're required to and even then encourage your major/minor to remove it from their curriculum. Beyond that its just a slog to get through this course.
- To a student who is interested in taking this course in the future I would recommend them to be active about their own education and focus on mastering topics early before it gets too late.
- This course is a lot more interesting than it sounds. It's not an easy course, but also isn't the total nightmare that I was anticipating. We learn a lot of math, but also understand the why through applications. In other words, we don't just do calculations, we apply the math and understand the mathematical concepts. The applications make things more interesting and meaningful. My main advice is to read the textbook and take all assignments seriously. (It would be hard to catch up if you get behind by skimping on an assignment). Don't be afraid to speak up to ask questions.

Recommended books, and other resources

BOOKS. *The Feynman Lectures on Physics, Volume II* (assigned for the vector-calculus unit; see above). Boas, *Mathematical Methods in the Physical Sciences*: the classic alternative presentation – if Felder's treatment of a topic isn't clicking, try Boas's; I taught from her book for a decade and am happy to point you at the parallel section. Schey, *div grad curl and all that*: an extremely conversational introduction to (and refresher on) vector calculus. Boyce and DiPrima, *Elementary Differential Equations and Boundary Value Problems*: a thorough reference for more depth on differential equations. Arfken and Weber, *Mathematical Methods for Physicists*: an encyclopedic reference; too dry to learn from in this course, but useful later in your career.

PEOPLE AND PLACES. There are many resources at Smith to help you succeed in this course and in your college career:

- Your instructor! I am here to help you. That's my job, so don't think it's bothering me to have you reach out with questions or concerns.
- Our course tutors, Aimen and Addie.[11]
- The Spinelli Center – open hours, one-on-one appointments, and math review workshops.
- Writing help, learning-specialist services, and workshops on time management and managing stress, all from the Jacobson Center.
- Crisis resources, counseling, and wellness services at the Schacht Center.

[11] **[TODO: tutoring hours and location, once scheduled]**

- Resources for gender identity and expression. **[TODO: link]**
- Where to report sexual misconduct and other forms of discrimination. Please note that your instructors are Responsible Reporters.